NeuralRetail

AI-Powered Sales Intelligence & Predictive Analytics Platform

Demand Forecasting · Customer Intelligence · Churn Prediction · Revenue Optimization

≤ 10%	≥ 0.90	< 4 min	15M+ txns
MAPE Target	AUC-ROC	Processing	Monthly scale

PROJECT CODE	AMX-DS-2026-04	VERSION	1.0 – Industry Edition
DOMAIN	Data Science & Analytics	DATE	April 2026
PREPARED FOR	Amdox – Engineering Division	CLASSIFICATION	Internal / Portfolio

1 Executive Overview

Mission statement

NeuralRetail is an end-to-end AI-powered sales intelligence platform designed for Amdox's enterprise clients in retail, FMCG, and e-commerce. The platform ingests multi-source transactional, behavioural, and external data to produce accurate demand forecasts, actionable customer intelligence, churn predictions, and inventory optimization recommendations — all served through an interactive analytics dashboard and REST API.

Why NeuralRetail matters for Amdox Technologies

Retailers globally lose an estimated $1.75 trillion annually from stockouts and overstocking. NeuralRetail positions Amdox as a full-stack data intelligence partner, delivering 30–50% stockout reduction, 15–25% revenue uplift, and 20–35% improvement in customer retention. The platform's MLOps backbone ensures models stay accurate over time without manual intervention — a genuine competitive differentiator.

1.1 Business impact targets

Business Objective	Metric	Target	Measurement Method
Stockout Reduction	Stockout frequency rate	↓ 30–50%	Weekly inventory audit
Revenue Uplift	GMV per active SKU	↑ 15–25%	A/B test vs baseline
Customer Retention	30-day churn rate	↓ 20–35%	Cohort retention analysis

Business Objective	Metric	Target	Measurement Method
Forecast Accuracy	MAPE (30-day horizon)	≤ 10%	MLflow experiment log
Processing Throughput	Daily batch job runtime	< 4 min / 15M txns	Airflow task duration
Model Reliability	Prediction drift (PSI)	< 0.2	Evidently AI report
Dashboard Response	P95 API latency	< 1.5 seconds	Prometheus metrics

1.2 Target users & use cases

User Segment	Primary Role	Key Use Cases
Retail Category Managers	Strategic planning	Demand forecasts, seasonal trend analysis, assortment decisions
Supply Chain Analysts	Operational execution	Reorder automation, supplier lead-time adjustment, stockout alerts
CRM / Marketing Teams	Customer engagement	Churn risk lists, segment campaigns, RFM-based personalization
Finance Controllers	Budget & margin control	Revenue projections, overstock write-off reduction, margin analysis
C-Suite / Executives	Performance oversight	KPI dashboards, predictive P&L, market share intelligence
Data / ML Engineers	Platform operation	MLflow tracking, retraining pipeline, model drift monitoring

1.3 Non-functional requirements

NFR Category	Requirement	Target	Validation Method
Model Accuracy	Demand MAPE	≤ 10% (30-day)	MLflow + hold-out test set
Churn AUC-ROC	Classification quality	≥ 0.90	Stratified k-fold CV
Throughput	Daily batch processing	15M+ txns < 4 min	Airflow DAG timing
Scalability	Horizontal scale	Auto-scale on >70% CPU	Kubernetes HPA stress test
Observability	Drift detection coverage	100% of production models	Evidently AI dashboard
Availability	Dashboard uptime	99.5% monthly	UptimeRobot monitoring
Reproducibility	Experiment tracking	All runs logged & versioned	MLflow Model Registry audit
Data Privacy	PII handling	Anonymized at ingestion; GDPR Art. 17	Data audit log review

2 Core Functional Requirements

The platform delivers eight production-grade analytical capabilities. Each requirement includes detailed acceptance criteria and measurable production metrics to ensure delivery quality and business alignment.

ID	Capability	Detailed Description & Business Value	Acceptance Criteria & Metrics
F-01	Multi-Source Data Ingestion & ETL	Ingest POS transactions, e-commerce events, ERP inventory feeds, weather/macro signals, and competitor pricing via batch (CSV/Parquet/JDBC) and streaming (Kafka) connectors. Great Expectations data quality gates on every run.	Zero silent failures; DQ score ≥ 98%; SLA < 4 min; full lineage in DataHub
F-02	Advanced Customer Segmentation	RFM scoring + behavioural clustering using K-Means, DBSCAN, and Gaussian Mixture Models. Silhouette-optimized cluster count (6–10 segments). Each segment labelled with actionable business persona.	Silhouette score ≥ 0.55; 6–10 interpretable segments; stability ≥ 80% week-on-week
F-03	Demand Forecasting Engine	Hybrid Prophet + LSTM (PyTorch Lightning) ensemble for SKU-level forecasting at daily, weekly, and monthly granularity. External regressors: promotions, holidays, weather. Probabilistic intervals for safety-stock.	MAPE ≤ 10% on 30-day horizon; 90% PI coverage ≥ 88%; weekly retrain; MLflow logged
F-04	Churn Prediction & Retention Engine	XGBoost + LightGBM stacked classifier with SHAP explainability. Customer-level churn scores with actionable retention recommendations. High-risk cohort auto-export to CRM.	AUC-ROC ≥ 0.90; Precision@Top20% ≥ 0.78; SHAP waterfall per prediction; weekly batch
F-05	Revenue & Price Intelligence	Price elasticity modelling via DoWhy causal inference. Competitor price integration. Promotion lift analysis with pre/post attribution. What-if revenue simulator.	Elasticity R^2 ≥ 0.72; simulator response < 2s; attribution confidence ≥ 90%
F-06	Inventory Optimization & Reorder	EOQ + safety-stock driven by forecasted demand and lead-time distributions. Automated PO recommendations. Dead-stock risk scoring. ABC-XYZ classification.	Overstock/understock ↓ 30%; PO draft < 30s; dead-stock flag accuracy ≥ 85%
F-07	Interactive Analytics Dashboard	Streamlit 5-page app: executive KPI, demand explorer, customer hub, churn heatmap, inventory health. Drill-down to SKU/customer level. Export to PDF and Excel. Role-based access.	Lighthouse ≥ 85; P95 load < 1.5s; exports < 10s; fully mobile-responsive
F-08	MLOps & Automated Retraining	Airflow DAGs for training, validation, and promotion. Evidently AI drift reports. Auto-retrain on PSI > 0.2 or MAPE degradation > 15%. Champion/challenger A/B via MLflow Registry.	Drift detected < 24h; retrain < 20 min; zero downtime swap; full audit trail

3 Production Technology Stack – 2026

Every technology selection was evaluated against at least two alternatives, balancing ecosystem maturity, performance, team familiarity, and total cost of ownership for a production ML platform targeting enterprise retail clients.

Layer	Primary Technology	Version	Rationale / Alternatives Evaluated
Language	Python	3.12	Dominant DS/ML ecosystem; Rust considered for ETL hot paths
Data Ingestion	Apache Spark + Delta Lake	3.5 / 3.0	Distributed ETL for 15M+ txns; Polars evaluated for small batches
Data Quality	Great Expectations + Soda Core	0.18 / 0.20	Automated DQ gates; dbt tests also integrated
Data Lineage	OpenLineage + Marquez	1.18	End-to-end dataset provenance; DataHub as alternative
Feature Store	Feast (Redis + S3 offline)	0.40	Consistent train/serve features; Tecton evaluated
Data Processing	Pandas + NumPy + Polars	2.2 / 1.26 / 0.20	Pandas for EDA; Polars for 10x faster batch transforms
ML Framework	Scikit-learn + XGBoost + LightGBM	1.4 / 2.1 / 4.3	Tree ensembles for tabular; CatBoost evaluated
Deep Learning	PyTorch Lightning + TorchForecast	2.3 / 1.4	LSTM/TFT models; TensorFlow evaluated
Forecasting	Prophet + NeuralProphet + TimeGPT	1.1 / 0.9 / API	Hybrid: statistical baseline + DL + foundation model
Explainability	SHAP + LIME + Captum	0.46 / 0.15 / 0.7	SHAP for tree models; Captum for PyTorch; LIME fallback
Causal Inference	DoWhy + EconML	0.11 / 0.15	Price elasticity & promotion attribution; CausalML evaluated
Experiment Tracking	MLflow	2.13	Experiments, artifact store, Model Registry; W&B evaluated
Hyperparameter Tuning	Optuna + Ray Tune	3.6 / 2.32	Optuna for single-node; Ray Tune for distributed HPO
Drift Detection	Evidently AI	0.4	Data + model drift reports; Nannyml evaluated
Pipeline Orchestration	Apache Airflow	2.9	DAG scheduling + retry + alerting; Prefect evaluated
Dashboard	Streamlit + Plotly	1.35 / 5.22	Rapid interactive analytics; Dash / Grafana evaluated
API Service	FastAPI + Uvicorn	0.111 / 0.29	Async REST for scoring; Flask evaluated
Database	PostgreSQL 17 + TimescaleDB	17 / 2.15	ACID + time-series; ClickHouse for analytics warehouse
Cache	Redis 8	8.0	Feature cache + session; Dragonfly evaluated
Containerization	Docker (Distroless base)	27	Minimal attack surface; Podman-compatible
Orchestration	Kubernetes + Helm 3	1.31	EKS/GKE deployment; ArgoCD GitOps delivery
CI/CD	GitHub Actions + ArgoCD	—	Matrix test → build → push → deploy pipeline
Monitoring	Prometheus + Grafana + Loki	2.53 / 11.1 / 3.0	Metrics + dashboards + logs; OpenTelemetry traces

Layer	Primary Technology	Version	Rationale / Alternatives Evaluated
Infrastructure	Terraform + Terragrunt	1.9 / 0.66	IaC for AWS / GCP multi-cloud; Pulumi evaluated
Cloud Target	AWS (primary) / GCP (secondary)	—	SageMaker for managed training; Vertex AI evaluated

4 Architecture Overview

4.1 MLOps pipeline architecture

The platform follows a Lakehouse + Feature Store + MLOps architecture with five distinct colour-coded layers: data ingestion, feature engineering, model training and registry, serving and dashboard, and monitoring with automated feedback.

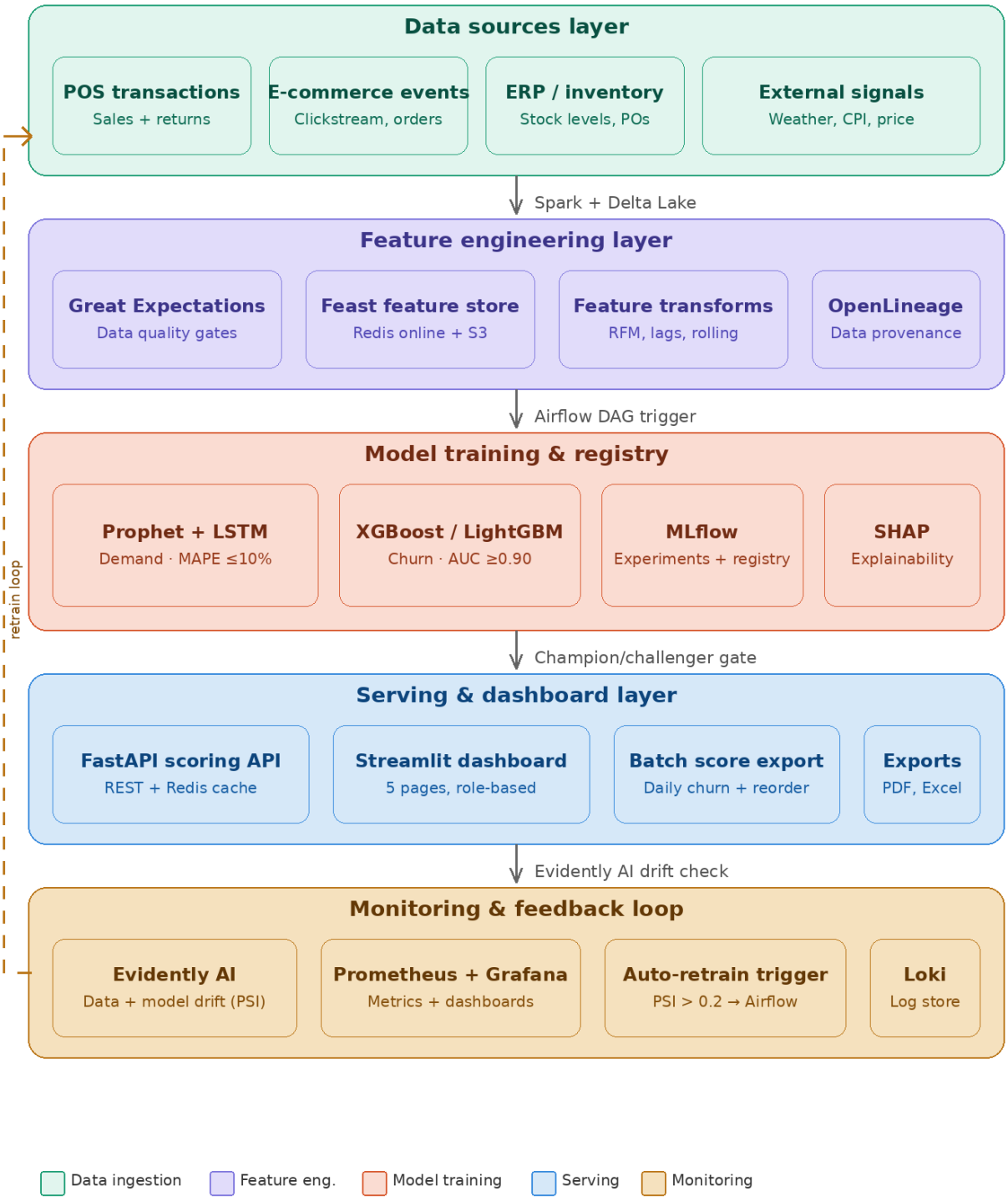

Figure 4.1 – NeuralRetail MLOps Pipeline Architecture | Amdox Technologies | April 2026

4.1a Layer reference

Layer	Colour	Key Components	Output / Artefact
Data Sources	Teal	POS transactions, e-commerce events, ERP feeds, weather API, competitor pricing	Raw data partitions in Delta Lake bronze layer
Feature Engineering	Purple	Great Expectations DQ gates, Feast feature store (Redis + S3), Polars transforms, OpenLineage lineage	Validated feature views; point-in-time training datasets
Model Training & Registry	Coral	Prophet + LSTM ensemble, XGBoost/LightGBM churn, MLflow experiments, SHAP, Optuna HPO	Registered models in MLflow; champion/challenger gates
Serving & Dashboard	Blue	FastAPI REST scoring API, Streamlit 5-page dashboard, batch score export, PDF/Excel reports	Real-time predictions; live KPI panels; exportable reports
Monitoring & Feedback	Amber	Evidently AI drift (PSI), Prometheus + Grafana metrics, Loki logs, auto-retrain Airflow trigger	HTML drift reports; retrain DAG on PSI > 0.2; PagerDuty alerts

4.2 Data flow patterns

Pattern	Components Involved	Trigger	Output Artefact
Batch ETL + DQ	Spark → Delta Lake → Great Expectations	Airflow daily 02:00 UTC	Validated Parquet partitions + DQ report
Feature Materialisation	Feast → Redis (online) + S3 (offline)	Post-ETL DAG step	Versioned feature views in Feature Store
Forecasting Training	Feature Store → Prophet/LSTM → MLflow	Weekly Sunday 03:00 UTC	Registered model in MLflow with metrics
Churn Scoring Batch	Feature Store → XGBoost → PostgreSQL	Daily 05:00 UTC	customer_churn_scores table + Airflow alert
Drift Check	Production predictions → Evidently AI	Daily post-scoring	HTML drift report + PSI metric to Prometheus
Auto-Retrain Trigger	PSI > 0.2 or MAPE ↑ > 15% → Airflow	Drift alert webhook	New challenger model registered in MLflow
Dashboard Refresh	PostgreSQL + Redis → Streamlit + FastAPI	On-demand + SSE push	Live KPI panels, charts, export files

4.3 Model selection & evaluation framework

Model / Task	Algorithm	Primary Metric	Threshold	Explainability
Demand Forecasting	Prophet + LSTM Ensemble	MAPE, RMSE, 90%-PI Coverage	MAPE ≤ 10%	Component decomposition + residuals
Churn Classification	XGBoost + LightGBM Stack	AUC-ROC, Precision@20%, F1	AUC ≥ 0.90	SHAP waterfall + beeswarm plots
Customer Segmentation	K-Means + DBSCAN + GMM	Silhouette, Davies-Bouldin	Silhouette ≥ 0.55	Cluster centroid profiles + radar charts
Price Elasticity	OLS + Ridge + DoWhy IDA	R^2, RMSE, Causal Estimand CI	$R^2 \geq 0.72$	Elasticity coefficient + sensitivity
Inventory EOQ	Analytical EOQ + Simulation	Stockout rate, holding cost	Stockout ↓ 30%	Parameter sensitivity + scenario table

Model / Task	Algorithm	Primary Metric	Threshold	Explainability
Revenue Forecasting	LightGBM + Prophet Hybrid	MAPE, directional accuracy	MAPE ≤ 8%	Feature importance + partial dependence

5 Granular 28-Day Execution Plan

Week 1 – Data discovery, ingestion & feature engineering

Day 1 – Project setup & data discovery

- Stakeholder requirements workshop: define KPI targets, data sources, and success criteria
- Competitor analysis: Databricks Retail AI, Google Cloud Retail, AWS Forecast, Azure Synapse
- Dataset acquisition: Kaggle RetailRocket, M5 Forecasting Competition, UCI Online Retail II
- Monorepo setup: Python + Poetry + Ruff + Black + pre-commit hooks + conventional commits
- Docker Compose local stack: PostgreSQL 17, Redis 8, MLflow, Airflow, Streamlit

Day 2 – Data ingestion pipeline

- Spark ingestion job: read CSV/Parquet sources → Delta Lake bronze layer with schema enforcement
- Great Expectations suite: define expectations on row counts, nullability, value ranges, foreign keys
- OpenLineage metadata emission to local Marquez instance for lineage tracking
- Airflow DAG skeleton: ingest_bronze → validate_dq → notify_on_failure
- Unit tests for ingestion functions; mock S3 with moto for CI compatibility

Day 3 – EDA & feature engineering

- EDA notebook: distribution analysis, outlier detection, correlation heatmap, seasonality decomposition
- Feature engineering: RFM scores, rolling mean/std demand (7/14/30 days), lag features (t-1, t-7, t-14)
- Date features: day-of-week, week-of-year, holiday flags, promotional event flags
- External data join: weather (Open-Meteo API), macro CPI index by product category
- Silver layer Delta table: clean, enriched, feature-ready dataset with data versioning

Day 4 – Feature store & data quality

- Feast feature definitions: customer, product, and transaction feature views
- Online store materialization to Redis for low-latency serving at inference time
- Offline store to S3 Parquet for training dataset retrieval with point-in-time joins
- Great Expectations checkpoint integration into Airflow with Slack alerting on DQ failures
- Data profiling report: YData Profiling HTML export for stakeholder review

Day 5 – Baseline models & MLflow setup

- MLflow tracking server setup: PostgreSQL backend + S3 artifact store + experiment UI
- Baseline demand model: Prophet with default seasonality; log MAPE, RMSE to MLflow
- Baseline churn model: Logistic Regression; log AUC-ROC, F1, confusion matrix to MLflow
- MLflow Model Registry: register first baseline models as 'Staging' for version tracking
- Experiment comparison dashboard in MLflow UI; document baseline as benchmark

Day 6 – Time-series deep dive

- Stationarity tests: ADF, KPSS; differencing and log-transform where needed
- STL decomposition: trend, seasonality, residual analysis for top-20 revenue SKUs
- Autocorrelation and PACF analysis to inform ARIMA order selection
- Multi-seasonality detection: weekly + annual patterns; Prophet with dual seasonality
- Hyperparameter grid search for Prophet: changepoint_prior_scale, seasonality_mode

Day 7 – Week 1 checkpoint

- Deliverables: EDA HTML report, cleaned Delta Lake silver table, baseline models in MLflow
- Internal review: data quality scores, baseline MAPE, AUC-ROC benchmarks vs targets
- ADR-001: Feature store strategy (Feast vs inline computation) documented in Notion
- Update sprint board; resolve any data access blockers; Week 2 plan confirmed

Week 2 – Advanced ML models & ensemble construction

Day 8 – LSTM forecasting model

- PyTorch Lightning LSTM: window-based input (28-day lookback), multi-step output (1/7/30 day)
- TorchForecast TFT (Temporal Fusion Transformer) as secondary deep learning architecture
- Training loop: gradient clipping, learning rate scheduler (CosineAnnealingWarmRestarts)
- Probabilistic output: quantile regression (q10, q50, q90) for safety-stock interval estimation
- MLflow autolog: hyperparameters, training curves, validation MAPE per forecast horizon

Day 9 – Prophet + LSTM ensemble

- Ensemble strategy: weighted average with weights optimized by Optuna on validation set
- NeuralProphet as intermediate model bridging statistical and deep learning approaches
- TimeGPT zero-shot forecast as benchmark comparison (API call, no fine-tuning required)
- Final ensemble MAPE target: ≤ 10% on 30-day horizon across all ABC-classified SKU tiers
- Confidence interval calibration: isotonic regression on quantile outputs for reliability

Day 10 – Churn prediction model

- Feature set: RFM metrics, purchase frequency decay, category switching signals, recency cliff
- XGBoost: Bayesian hyperparameter optimization with Optuna (100 trials, 5-fold stratified CV)
- LightGBM: DART boosting mode for regularization; benchmark vs XGBoost on AUC-ROC
- Stacking: meta-learner (Logistic Regression) on XGB + LGBM + RF out-of-fold predictions
- Class imbalance: SMOTE-NC oversampling + scale_pos_weight; evaluate at threshold=0.35

Day 11 – SHAP explainability & model interpretation

- SHAP TreeExplainer for XGBoost churn model: global feature importance + local explanations
- SHAP beeswarm, waterfall, and dependence plots for top-10 predictive features
- LIME explanations as secondary interpretability check for model robustness validation
- Captum integrated gradients for LSTM temporal feature attribution visualization
- Business narrative: translate SHAP values into plain-language retention action recommendations

Day 12 – Customer segmentation & RFM intelligence

- K-Means: Elbow + Silhouette analysis (k=6–10); final k selected by Davies-Bouldin index
- DBSCAN: auto eps via k-distance graph; identify outlier / ultra-VIP customer clusters
- Gaussian Mixture Models: probabilistic segment membership for borderline customers
- Segment profiling: revenue contribution, churn risk, category affinity, CLV estimate per cluster
- CRM export: high-churn-risk segment with SHAP-based retention action codes

Day 13 – Price elasticity & revenue intelligence

- Log-log OLS regression: price vs demand per category with promotion dummy variables
- DoWhy causal graph: identify and adjust for confounders (season, competitor price, ad spend)
- EconML Double ML for heterogeneous treatment effects of price changes across segments
- Promotion lift analysis: pre/post attribution with synthetic control method
- What-if revenue simulator: adjust price/promotion inputs → see projected demand and revenue impact

Day 14 – Week 2 checkpoint

- Deliverables: ensemble forecast (MAPE ≤ 10%), churn model (AUC ≥ 0.90), segmentation, elasticity model
- MLflow Model Registry: promote best models from Staging → Production after validation gate
- Champion/challenger: new model must beat current champion by ≥ 5% on unseen hold-out set
- Weekly model report: MAPE by SKU tier, AUC by cohort, segment stability — logged to MLflow

Week 3 – Dashboard, API layer & MLOps pipeline

Day 15 – Streamlit dashboard architecture

- Multi-page layout: Executive Overview / Demand Intelligence / Customer Hub / Inventory / MLOps Monitor
- Authentication: Streamlit-Authenticator with role-based page access (Admin, Analyst, Viewer roles)
- Navigation: session state management, query parameter routing, sharable bookmark support
- Amdox brand design system: orange palette (#E84E1B, #F7941D, #FBBA13), custom CSS, responsive layout
- Plotly charts: interactive time-series, heatmaps, scatter with trendlines, Sankey diagrams

Day 16 – Demand forecast dashboard module

- SKU-level forecast explorer: date range picker, category/region filter, SKU typeahead search
- Forecast chart: actual vs predicted with confidence interval bands (Plotly filled area)
- What-if simulator: adjust promotion flag / weather / price coefficient → see demand impact live
- Forecast accuracy leaderboard: MAPE ranking table by SKU, category, and store with trend
- Export: forecast table to Excel, chart to PNG, full report to PDF (ReportLab generation)

Day 17 – Customer intelligence & churn module

- Churn risk heatmap: customer segments × risk decile matrix with click-through drill-down
- Individual customer 360 view: SHAP waterfall chart, purchase timeline, segment membership badge
- Segment comparison: radar chart of RFM dimensions across all clusters with toggle overlay
- Retention action board: auto-generated plain-language recommendations per churn risk tier
- Campaign builder UI: one-click CRM CSV export of selected risk segment with action codes

Day 18 – Inventory optimization module

- Inventory health scorecard: ABC-XYZ classification matrix, stockout risk by SKU, dead-stock flags
- EOQ calculator UI: input lead time and holding cost → recommended reorder quantity + safety stock
- Reorder alert table: SKUs below safety stock threshold with urgency scoring and supplier info
- Overstock risk panel: SKUs projected to exceed capacity + liquidation/markdown recommendations
- Supplier lead-time tracker: historical variance analysis → buffer stock adjustment suggestions

Day 19 – FastAPI scoring API

- REST endpoints: /predict/demand, /predict/churn, /segment/score, /inventory/reorder, /health
- Request/response validation: Pydantic v2 schemas with strict type enforcement and field-level docs
- Model loading from MLflow production URI: no code dependency on training environment
- Redis caching: cache SKU-level forecasts for 1 hour to reduce redundant model inference calls
- Auth: API key + JWT bearer tokens; sliding-window rate limiting: 100 req/min per client key

Day 20 – Evidently AI drift monitoring

- Data drift report: PSI and Kolmogorov-Smirnov test on all input features vs training distribution
- Model performance monitoring: MAPE tracking over rolling 7-day window with trend alerting
- Prediction drift: Jensen-Shannon divergence on forecast output distributions
- Automated report: HTML saved to S3 + PSI/MAPE summary metrics exported to Prometheus
- Alerting: Airflow sensor polls Prometheus; triggers retraining DAG automatically if PSI > 0.2

Day 21 – Week 3 checkpoint

- Deliverables: 5-page Streamlit dashboard, FastAPI scoring service, Evidently AI monitoring setup
- End-to-end smoke test: ingest → feature → train → predict → dashboard display → export
- Performance: Lighthouse score ≥ 85 on dashboard; API P95 latency < 1.5s under load
- Security scan: Trivy on all Docker images; Bandit static analysis on Python source code

Week 4 – Deployment, production polish & delivery

Day 22 – Containerisation & Docker optimisation

- Multi-stage Dockerfiles: builder stage (full deps) → runtime stage (Distroless Python)
- Separate images: airflow-worker, mlflow-server, streamlit-app, fastapi-scorer
- docker-compose.prod.yml: health checks, resource limits, named volumes, restart policies
- Container security: non-root user, read-only filesystem, dropped Linux capabilities (CAP_DROP ALL)
- Image size optimisation: target < 500 MB per service via .dockerignore and layer caching

Day 23 – Kubernetes & Helm deployment

- Helm chart: values.yaml + templates for Deployment, Service, Ingress, HPA, PodDisruptionBudget
- Kubernetes security: NetworkPolicy (deny-all + selective allow), ResourceQuota per namespace
- Horizontal Pod Autoscaler: scale FastAPI on >70% CPU; Streamlit on >60% memory utilization
- Sealed Secrets for all credentials; ConfigMap for non-sensitive environment configuration
- minikube / kind local cluster validation; ArgoCD Application manifest committed to GitOps repo

Day 24 – CI/CD pipeline

- GitHub Actions matrix: lint (ruff/black) → unit tests (pytest) → integration tests → Docker build
- Model validation gate in CI: runs hold-out evaluation; fails if MAPE > 12% or AUC < 0.88
- Docker push to ECR/GCR; image tagged with git SHA + semantic version for traceability
- ArgoCD GitOps: staging auto-deploy on merge to main; production requires manual approval gate
- Slack notifications: pipeline status + model metric summary posted on each run completion

Day 25 – Cloud deployment

- Frontend: Streamlit on AWS EC2 / Streamlit Community Cloud with Cloudflare HTTPS proxy
- API: FastAPI on AWS ECS Fargate / Railway / Fly.io with auto-scaling and zero-downtime deploys
- MLflow: managed on EC2 + RDS PostgreSQL backend + S3 artifact store with lifecycle policies
- Airflow: MWAA (Managed Airflow) or self-hosted on EKS with Celery executor and Redis broker
- Database: RDS Aurora Serverless v2 + ElastiCache Redis; custom domain + Let's Encrypt TLS

Day 26 – Observability stack

- Prometheus scrape configs: FastAPI /metrics, Airflow StatsD, model latency histograms
- Grafana dashboards: model performance over time, API latency P50/P95/P99, drift scores, SLA
- Loki log aggregation: structured JSON logs from all services with correlation ID propagation
- OpenTelemetry SDK: distributed traces from API request → feature fetch → model inference
- PagerDuty alerts: drift breach, API error rate > 1%, Airflow SLA miss, retraining failure

Day 27 – Demo video & documentation

- Record 5–8 min scenario-based demo: end-to-end from raw data ingestion to dashboard insights
- Demo scenarios: demand forecast for seasonal SKU, churn risk intervention, automated reorder trigger
- README.md: architecture diagram, docker-compose local setup, environment variables table, screenshots
- Jupyter notebooks: EDA walkthrough, model training with inline comments, evaluation analysis
- Model cards: training data summary, metrics, bias evaluation, intended use — one per registered model

Day 28 – Final QA, submission & personal reflection

- Cross-browser QA: Chrome, Firefox, Safari, Edge; mobile responsive validation at 375px / 768px
- Final Lighthouse run: Performance ≥ 85, Accessibility ≥ 90, Best Practices ≥ 90
- k6 load test: 200 concurrent virtual users on dashboard + API; validate P95 < 1.5s sustained
- Submission package: PDF report + GitHub repo + live HTTPS demo URL + video link
- Personal Reflection: key learnings, MLOps challenges overcome, future roadmap ideas documented

6 Security, Ethics & Data Privacy

Domain	Controls Implemented	Standard / Tool
Data Anonymization	PII masked at ingestion (customer_id SHA-256 hashed, email/phone scrubbed); k-anonymity check on exports	GDPR Art. 5, 25
Access Control	Role-based dashboard access; API key + JWT for scoring; HashiCorp Vault for secret management	NIST 800-53 AC-6
Input Validation	Pydantic v2 strict schemas on all API inputs; SQLAlchemy ORM prevents SQL injection	OWASP A03:2021
Model Fairness	Bias audit: churn model evaluated across demographic proxies (region, age group); equal opportunity metric	EU AI Act Art. 10
Dependency Security	Snyk SCA scan in CI; pip-audit weekly; Renovate bot for automatic dependency version updates	CIS Controls v8
Secrets Management	HashiCorp Vault / AWS Secrets Manager; Sealed Secrets in Kubernetes; zero hardcoded secrets policy	CIS K8s Benchmark
Audit Logging	Immutable log: all model predictions + dashboard exports timestamped and hash-chained	SOC 2 CC7.2
Container Security	Trivy scan in CI; Distroless base images; non-root user; read-only filesystem; CAP_DROP ALL	CIS Docker Benchmark

7 MLOps Maturity & Best Practices

MLOps Dimension	Practice Applied	Tools
Reproducibility	Random seed fixed, data version pinned, environment locked via requirements.txt + Poetry lock	MLflow, DVC
Data Versioning	Delta Lake time-travel; dataset snapshots versioned in DVC; hash-based artifact naming in MLflow	Delta Lake, DVC
Model Versioning	Semantic versioning in MLflow Model Registry; champion/challenger stage gates with metrics threshold	MLflow Registry
Automated Retraining	Airflow DAG triggered by Evidently AI drift alert (PSI > 0.2) or scheduled weekly Sunday 03:00 UTC	Airflow, Evidently AI
A/B Deployment	Shadow mode: new model runs alongside champion; promote only if ≥ 5% improvement on hold-out set	MLflow, Istio
Monitoring	Data drift, model performance drift, and prediction distribution tracked daily with auto-alerting	Evidently AI, Prometheus
Testing	Unit tests for feature functions; integration tests for Airflow DAGs; model regression tests in CI gate	pytest, Great Expectations
Documentation	Model cards per registered model: training data, metrics, bias evaluation, intended use, limitations	MLflow, Notion

8 Evaluation Criteria (100-Point Scale)

Category	Points	Focus Areas
Innovation & Business Problem Solving	15	Original analytical approaches, quantified KPIs, thoughtful feature engineering decisions
Technical Depth & Model Quality	25	Algorithm selection rigor, ensemble design, hyperparameter optimization, evaluation methodology
MLOps & Production Readiness	20	Drift detection, automated retraining, MLflow tracking, CI/CD, reproducible pipelines
Documentation Quality	20	Professional PDF report, architecture diagrams, model cards, README depth and clarity
Deployment & Live Demo Reliability	10	Stable HTTPS demo, fast load, functional without sign-up, k6 load test evidence
Presentation & Visual Polish	10	Demo video quality, dashboard UX, chart clarity, storytelling effectiveness
TOTAL	100	

9 Submission Guidelines

#	Deliverable	Format / Location	Required?	Weight
1	Project Report (this document)	1 PDF, A4, 10–18 pages	Yes	25%
2	Live Public Demo URL	HTTPS (Streamlit Cloud / Hugging Face / AWS)	Yes	30%
3	GitHub Repository	Public repo with notebooks, src/, tests/, Dockerfile	Yes	20%
4	README.md	Setup guide, architecture, screenshots, video link, model metrics	Yes	15%
5	Demo Video	5–8 min, YouTube Unlisted / Loom / Google Drive	Yes	10%

Naming convention: YourName_NeuralRetail_AmdoxDS_April2026.pdf / .zip

Submission: Via the Amdox Technologies project dashboard only — no email submissions. Late submissions are automatically disqualified from stipend consideration.

File size limits: ZIP ≤ 500 MB | Individual PDFs ≤ 12 MB | No private API keys required to run demo.

Stand-out practices (highly recommended)

- Include SHAP summary plots, drift simulation screenshots, and model comparison tables in the report
- Record a live drift detection + auto-retraining demonstration in the demo video
- Add a Model Card for each registered model in MLflow (training data, metrics, bias evaluation)

- Show Evidently AI HTML drift report as a screenshot or embedded link in README
- Document MLOps maturity level: describe how your pipeline would scale to 100x data volume
- End the report with a Personal Reflection: key learnings, challenges overcome, future roadmap

Crafted with precision and modern data science principles

Amdox Technologies · Data Science & Analytics Domain · April 2026